Title: Guilty vs Not Guilty Classification from Old Bailey Trial Texts using NLP Techniques

Group -3

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1. Background and Motivation

The Proceedings of the Old Bailey provide a large collection of digitised criminal trial accounts from London's central criminal court (1674–1913), with both narrative trial text and structured metadata such as date, offence, verdict, and punishment. The Old Bailey project provides access to the underlying XML data, enabling researchers to extract and analyse trial records programmatically.

This project focuses on building an end-to-end, basic NLP pipeline—from extraction and cleaning to classical machine learning classification—to predict guilty vs not guilty outcomes using only the trial text.

2. Literature Review

Legal NLP research commonly frames tasks like document classification, information extraction, and judgment/outcome prediction as supervised learning problems over legal text, but highlights practical issues such as long documents, domain-specific language, limited open datasets, and the need for interpretability. In the specific area of Legal Judgment Prediction (LJP), a survey by Cui et al. reviews many datasets, evaluation metrics, and baseline-to-neural modelling approaches, and defines LJP as predicting case outcomes from textual fact descriptions.

For the Old Bailey corpus, the official data documentation notes that the XML is "large and complex," and that a single trial may contain multiple defendants, offences, victims, verdicts, and sentences, which motivates careful preprocessing and label construction. Digital humanities work also emphasises that Old Bailey trials are mediated historical texts whose markup and conventions vary across time, affecting extraction and analysis.

Given these considerations, this project adopts interpretable classical ML baselines (e.g., TF–IDF + logistic regression / Naive Bayes / linear SVM) to create a transparent benchmark for guilty vs not guilty prediction on historical trial narratives.

3. Problem Statement and Objectives

Problem statement:

Given the textual record of a trial from the DHI Old Bailey dataset, predict whether the final verdict is guilty or not guilty, using only the trial text (not the verdict field) as model input.

Objectives:

- Extract trials and verdict labels from the DHI Old Bailey data and construct a binary classification dataset (guilty vs non-guilty).
- Clean and preprocess the trial text (remove markup, normalise spacing and punctuation, handle basic historical spelling issues).
- Represent the text using basic NLP methods (e.g., bag-of-words, TF–IDF, simple n-grams).
- Train and evaluate standard machine learning classifiers (e.g., logistic regression, Naive Bayes, linear SVM) on the classification task.
- Analyse which words or phrases are most associated with guilty vs not guilty outcomes, to gain interpretable insights into language use in the proceedings.

4. Dataset Description

The Old Bailey Proceedings Online project provides a fully digitised, structured corpus of all surviving published trial accounts from 1674–1913, including text and metadata. The "Access to Old Bailey Online Data" documentation explains that each data file corresponds to a court session and contains multiple trial records with XML tags for key fields such as defendant, offence, verdict, and sentence.

For this project, the dataset will be obtained from the DHI Old Bailey data page (https://www.dhi.ac.uk/data/oldbailey), where researchers can download the underlying trial data for non-commercial research. From each trial record, the project will use:

- **Text:** main trial narrative (testimony, examination, statements).
- **Label:** verdict mapped into binary classes (guilty vs non-guilty).
- **Optional metadata:** year and offence category for simple descriptive statistics and stratified train–test splits.

5. Methodology

5.1 Data Extraction and Cleaning

- Parse the downloaded Old Bailey data files (e.g., XML) to extract trial-level text and verdict for each case.
- Remove markup tags, page headers, and boilerplate text; keep only trial narrative segments.
- Basic text cleaning: lowercasing, removing extra whitespace, standardising punctuation, and optionally removing numbers and very rare characters.
- Simple handling of historical spelling variants (e.g., mapping some frequent archaic forms to modern equivalents using a small custom dictionary) to reduce sparsity, while keeping the approach easy to implement.

5.2 NLP Preprocessing and Feature Engineering

Using standard NLP libraries (e.g., NLTK, spaCy or scikit-learn text utilities), the project will:

- Tokenise the text into words.
- Optionally remove common English stopwords; experiment with versions with and without stopword removal.
- Apply stemming or lemmatisation to reduce inflected forms.
- Build:
    - Bag-of-words representations (count vectors).
    - TF–IDF vectors with unigrams and possibly bigrams, which are widely used baselines for legal text classification and summarisation.

5.3 Classification Models

The study will focus on simple, interpretable classifiers:

- Multinomial Naive Bayes.
- Logistic regression with L2 regularisation.
- Linear Support Vector Machine (SVM).

These models are well-suited for high-dimensional sparse text features and are commonly used in baseline legal text classification tasks. The dataset will be split into training, validation, and test sets with stratification to preserve the guilty/non-guilty ratio.

Evaluation metrics will include accuracy, precision, recall, and F1 score. Because the distribution of verdicts in the Old Bailey is skewed towards guilty, macro-averaged F1 will be used to evaluate performance on both classes.

5.4 Analysis and Interpretation

- Inspect the most informative features (words or bigrams) for each class using model coefficients from logistic regression or SVM, and log-probabilities from Naive Bayes.
- Compare performance across offence categories or time periods to see whether models generalise consistently.
- Provide qualitative examples of trials that are misclassified and discuss potential reasons (e.g., unusual case type, very short text, ambiguous language).

This analysis connects to prior work showing that specific lexical and syntactic patterns can influence perceived blame and verdicts, even outside historical corpora.

6. Conclusion

This revised proposal presents a practical plan to build a complete, interpretable NLP pipeline for guilty vs not guilty prediction using the DHI Old Bailey trial texts. The added literature review situates the work within established legal NLP and LJP research, while also acknowledging Old Bailey-specific data complexities (XML structure, multiple entities per trial, and mediated historical text). The expected outcome is a reproducible dataset construction workflow, baseline classification results, and interpretable linguistic signals associated with verdict outcomes.

7. References

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